Use multiplication and division before addition and subtractio

Use the scaffold, show each step and ask for an adult check when marked.

Name: ______________________________ Date: ____________

1. Read the model: Calculate 18 + 6 x 7. Copy or complete the first step.

2. Calculate 90 - 36 / 6.

3. Calculate 120 / 5 + 8 x 3.

4. Miro says: Miro adds 18 and 6 before multiplying by 7. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Calculate 18 + 6 x 7. Copy or complete the first step.
   Answer 60.

2. Calculate 90 - 36 / 6.
   Answer 84.

3. Calculate 120 / 5 + 8 x 3.
   Answer 48.

4. Miro says: Miro adds 18 and 6 before multiplying by 7. Is this correct? Explain.
   Answer Multiplication has priority, so calculate 6 x 7 first.

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Work independently. Show a complete method and check at least one answer.

Name: ____ Date: __________________

1. Calculate 18 + 6 x 7.

2. Calculate 90 - 36 / 6.

3. Calculate 120 / 5 + 8 x 3.

4. Solve this independently and show every stage: Calculate 18 + 6 x 7.

5. Use a second method or representation for: Calculate 90 - 36 / 6.

6. Check this result using an inverse, estimate or known fact: Calculate 120 / 5 + 8 x 3.

7. Use the digits 2, 4 and 8 once to make an expression with answer 34.

8. Identify and correct this misconception: Miro adds 18 and 6 before multiplying by 7.

9. Write a complete sentence explaining how you checked one answer.

10. Write and solve a similar question to: Calculate 120 / 5 + 8 x 3.

Teacher answers and checking prompts

1. Calculate 18 + 6 x 7.
   Answer 60.

2. Calculate 90 - 36 / 6.
   Answer 84.

3. Calculate 120 / 5 + 8 x 3.
   Answer 48.

4. Solve this independently and show every stage: Calculate 18 + 6 x 7.
   Answer 60.

5. Use a second method or representation for: Calculate 90 - 36 / 6.
   Answer Expected result: 84. Method or representation may vary.

6. Check this result using an inverse, estimate or known fact: Calculate 120 / 5 + 8 x 3.
   Answer Expected result: 48. A valid check must be shown.

7. Use the digits 2, 4 and 8 once to make an expression with answer 34.
   Answer One answer is 2 + 4 x 8 = 34.

8. Identify and correct this misconception: Miro adds 18 and 6 before multiplying by 7.
   Answer Multiplication has priority, so calculate 6 x 7 first.

9. Write a complete sentence explaining how you checked one answer.
   Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

10. Write and solve a similar question to: Calculate 120 / 5 + 8 x 3.
    Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

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Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: ____ Date: __________________

1. Use the digits 2, 4 and 8 once to make an expression with answer 34.

2. Prove the answer to this question, rather than only calculating it: Calculate 18 + 6 x 7.

3. Solve this in two different ways and compare them: Calculate 90 - 36 / 6.

4. Work backwards from the answer to reconstruct the question: 48.

5. Explain why this reasoning fails: Miro adds 18 and 6 before multiplying by 7.

6. Create a new example that tests the same idea as: Calculate 120 / 5 + 8 x 3.

7. Find a second method or representation. Compare its efficiency with your first method.

8. Write one always, sometimes or never statement about today's learning and justify it.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Use the digits 2, 4 and 8 once to make an expression with answer 34.
   Answer One answer is 2 + 4 x 8 = 34.

2. Prove the answer to this question, rather than only calculating it: Calculate 18 + 6 x 7.
   Answer Expected result: 60. Proof or justification must match the lesson structure.

3. Solve this in two different ways and compare them: Calculate 90 - 36 / 6.
   Answer Expected result: 84. Two valid methods and a comparison are required.

4. Work backwards from the answer to reconstruct the question: 48.
   Answer One valid reconstruction is: Calculate 120 / 5 + 8 x 3. Other equivalent questions are acceptable.

5. Explain why this reasoning fails: Miro adds 18 and 6 before multiplying by 7.
   Answer Multiplication has priority, so calculate 6 x 7 first.

6. Create a new example that tests the same idea as: Calculate 120 / 5 + 8 x 3.
   Answer Answers vary. The example and solution must preserve the mathematical structure.

7. Find a second method or representation. Compare its efficiency with your first method.
   Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

8. Write one always, sometimes or never statement about today's learning and justify it.
   Answer Answers vary. A valid example and counterexample should support the classification.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
   Answer Answers vary. The error must be mathematically relevant and the correction must be precise.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
    Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.